Ailton Rocha

Threat Hunter & Detection Engineer

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SUMMARY

Threat Hunter and Detection Engineer with 7 years at Vivo (Telefônica Brasil), protecting a 50K+ endpoint environment with Linux, Windows and cloud servers. I run hypothesis-driven threat hunting with MITRE ATT&CK and TaHiTI, and turn findings into production detections on CrowdStrike Falcon (LogScale/CQL and custom IOAs), Cortex XDR and Splunk, cutting false positives by 90%. Solid offensive foundation: a career on the offensive side since 2009 (reverse engineering, anti-cheat and anti-debug for online game servers), DCPT and 4 Hack The Box Pro Labs. Thinking like the attacker is what makes my hunting and Blue Team detections more effective. Creator of knowledge-rag, a local RAG MCP server with 300+ stars on GitHub.

SKILLS

Threat Hunting & Detection: hypothesis-driven threat hunting, Detection Engineering, MITRE ATT&CK, TaHiTI, correlation rules, IOA and IOC, false positive tuning, Purple Team, Breach and Attack Simulation (BAS).

Platforms: CrowdStrike Falcon, Falcon LogScale (CQL), Cortex XDR, Splunk (SPL), Wazuh, Sysmon, Trellix DLP, Trend Micro XDR, Panda Security, Darktrace, Elastic.

Incident Response & Forensics: DFIR, malware analysis, reverse engineering, memory forensics (Volatility), Velociraptor, Autopsy, FTK, Wireshark.

Offensive Security: Active Directory (Kerberos, delegation, NTLM relay, ADCS ESC1 to ESC14), BloodHound, Impacket, NetExec, Rubeus, Certipy, Mimikatz, pivoting, EDR evasion.

Threat Intelligence: CTI, brand monitoring, credential leaks, phishing and typosquatting, ransomware, executive reporting.

Automation & AI: Python, PowerShell, Bash, REST APIs, Model Context Protocol (MCP), RAG, autonomous and multi-agent LLM systems.

PROFESSIONAL EXPERIENCE

Threat Hunter & Detection Engineer (Cyber Security Consultant) **02/2025 - Present**
Vivo (Telefônica Brasil), São Paulo, Brazil

- Run hypothesis-driven threat hunting across a 50K+ endpoint environment with CrowdStrike Falcon and Falcon LogScale (CQL), mapping technique coverage in MITRE ATT&CK with the TaHiTI methodology.
- Develop detection rules and custom IOAs on CrowdStrike and Cortex XDR; tuning reduced false positives by 90%.
- Built autonomous AI (LLM) agents for security operations: the N1 triage agent for Threat Intelligence alerts closed 62 of 62 cases on day one, all confirmed in blind review.
- Produce the monthly Threat Intelligence bulletin for leadership: telecom, financial sector, data leaks, ransomware and government.
- Run Purple Team with Cymulate on both sides: I create Breach and Attack Simulation tests, map APT TTP coverage in MITRE ATT&CK v19 and evaluate security solutions against those scenarios.
- Hands-on experience evaluating, testing and running EDR and XDR in production: CrowdStrike Falcon and Identity Protection, Cortex XDR, Trend Micro XDR, Panda Security and Darktrace.
- Reverse engineer encrypted malicious binaries and stress-test Identity Protection, EDR, XDR and SIEM solutions.

Senior Cyber Security Analyst, Threat Hunting **07/2023 - 02/2025**
Vivo (Telefônica Brasil), São Paulo, Brazil

- Hunted advanced threats (APT, ransomware, spear phishing) with hypothesis-driven methodology and MITRE ATT&CK.
- Created detection rules on CrowdStrike Falcon, Splunk (SPL), Sysmon, Wazuh and Trend Micro.
- Automated hunting workflows with Python and PowerShell, reducing manual triage by 60%.
- Wrote playbooks, runbooks and the LogScale reference documentation used by SOC teams.

Cyber Security Analyst, CSIRT **04/2020 - 07/2023**
Vivo (Telefônica Brasil), São Paulo, Brazil

- Investigated and contained ransomware, APT, targeted phishing and web attacks (SQL injection, XSS, CSRF).
- Analyzed malware and network traffic to extract indicators of compromise (IOCs); monitored AV, EDR and SIEM.

- Investigated servers and VPN access and produced technical incident reports.

Network Analyst **09/2019 - 01/2020**

Logicalis (Banco Santander account), São Paulo, Brazil

- Cisco equipment configuration and LAN, WAN and wireless troubleshooting in a banking environment.

Network Infrastructure Analyst **06/2019 - 09/2019**

Central IT (São Paulo Municipal Department of Education), São Paulo, Brazil

- Operated 10,000 switches and 8,000 access points across 1,500 schools; SonicWall and pfSense firewalls; Zabbix monitoring.

Network Analyst (Intern) **03/2017 - 02/2019**

NIC.br, IX.br project, São Paulo, Brazil

- BGP configuration in a multi-vendor environment (Cisco, Brocade, Extreme, Huawei) at IX.br, Brazil's internet exchange point; LACP and bilateral VLANs.

Online Game Server Administrator & Reverse Engineer (self-employed) **01/2009 - 01/2014**

Break GunZ, online game server, São Paulo, Brazil

- Built and administered online game servers on Linux and Windows VPS with SQL Server databases.
- Developed an anti-cheat system and anti-debug and anti-tamper protections for 64-bit Windows binaries, with DLLs in C++ and Assembly.
- Reverse engineered client and server binaries (OllyDbg, Immunity Debugger, Process Hacker) to find and close the flaws cheaters exploited.
- Protected the web layer with WAF and DDoS mitigation on Cloudflare plus anti-SQL injection scripts in PHP.

PROJECTS

- **knowledge-rag** (creator and maintainer), github.com/lyonzin/knowledge-rag: 100% local RAG MCP server for AI agents, with hybrid search (BM25, semantic and RRF), cross-encoder reranking and 20 file formats. 300+ stars, 49 forks, 28K+ downloads on PyPI and npm, CI with 35+ checks across 9 OS and Python combinations.
- **LogonHoursRobot** (private repository): fully automated PowerShell robot that applies and audits logon-hour restrictions across multiple Active Directory domains, with delta processing, byte-level verification, SHA-256 forensic evidence, an HTML dashboard and granular rollback; paginated LDAP tested with 23K+ users.
- **X-SHELL** (private repository): red team framework for authorized operations, with reverse shell session management, artifact signing and MCP automation with AI agents (Claude Code, OpenAI and Cursor).
- **Open security-operations tools**: PowerShell profile with utility functions, XOR string encoder for Blue Team, Splunk Enterprise install automation scripts and a Kibana misconfiguration auditor.

CERTIFICATIONS

eCTHP, Certified Threat Hunting Professional (v2) | INE Security | ID 96700960	03/2024
eCIR, Certified Incident Responder | INE Security | ID 104767045	05/2024
GCSE, Guardicore Certified Segmentation Engineer | Akamai Guardicore | ID GCSE-171243106	12/2023
GCSA, Guardicore Certified Segmentation Administrator | Akamai Guardicore | ID GCSA-170342830	12/2023
DCPT, Desec Certified Penetration Tester | Desec Security | ID FVNS-QIJMH-OUVZ	12/2022
SCFE, Sec4US Certified Forensics Expert | Sec4US	10/2022
PDPE, Privacy and Data Protection Essentials | EXIN	02/2021

PRACTICAL LABS: HACK THE BOX PRO LABS

RastaLabs, red team in Active Directory with defense evasion	11/2026
Zephyr, multi-domain Active Directory, relay, trusts and pivoting | HTBCERT-D6337FFDC3	04/2026
Puppet, C2 operations, Active Directory and DevOps | HTBCERT-86470B4B5B	02/2026
Dante, corporate network pentest with pivoting | HTBCERT-AAB49A3E31	12/2025

ADDITIONAL TRAINING

Detection, forensics and response

Advanced Digital Forensics for Windows (GHADFW) | GoHacking 2025
Reverse Engineering Malware (EHREM) | GoHacking 2023
GHSOC, Security Operation Center Foundations | GoHacking 2023
Splunk Enterprise 9.0 System Administration | Splunk 2023
Splunk Cloud Administration | Splunk 2023
SCFE, Digital Forensics Advanced | Sec4US 2022
Active Directory Defense (GHADD) | GoHacking, no date

Offensive security
Active Directory Certificate Services (GHADCS) | GoHacking 2024
Active Directory Operations (EHADOP) | GoHacking 2023
Endpoint Protection Evasion (EHEPE) | GoHacking 2023
Active Directory Exploitation | Sec4US 2023
Web API Exploitation | Sec4US 2023
Web Hacking na Prática 2.0 | CrowSec 2023
Pentest Experience | Desec Security 2022
DCPT, Pentest Profissional | Desec Security 2022
Smart Recon para Pentest | Desec Security
Web Dev Security | Desec Security
Defense Evasion | Desec Security
Web Hacking 3.0 | HackingClub
Endpoint Protection Evasion | Hack The Box
Wi-Fi Hacking Enterprise 1.0 | Desec Security 2021

AI and networking
Formação em Inteligência Artificial | branas.io 2026
IPv6 | NIC.br 2018
CCNA (course) | no date

EDUCATION

Postgraduate, Applied AI Engineering | Anhanguera | in progress 2026
Postgraduate, Cyber Security | Instituto Daryus 2019 - 2021
Computer Networks (Technologist) | Faculdade Impacta 2016 - 2018

LANGUAGES

Portuguese: native. English: basic.